Agglomerative Hierarchical Clustering

From Pairwise Distances to Linkage, Dendrograms and Ward's Method

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Learning Roadmap

Stage	Central question
1. Recap and hierarchy	Why group at several levels?
2. Distance and linkage	Which clusters should merge?
3. Manual numerical	How do distances change after each merge?
4. Dendrogram	Where should we cut the hierarchy?
5. Derivations	Why do the linkage updates work?
6. Evaluation and uses	How do we validate and apply the result?

Key takeaway

Intuition $\longrightarrow$ computation $\longrightarrow$ mathematical justification.

1. Clustering: A Brief Recap

Unsupervised learning

- Observations have features, but no supplied target labels.
- We seek groups that are meaningful under a chosen similarity measure.
- A cluster is a modeling choice: representation, scaling and distance all matter.

Key takeaway

K-Means gives hard assignments; Fuzzy C-Means gives membership degrees. Hierarchical clustering gives nested groups.

Why a Hierarchy?

- Documents: science $\rightarrow$ computer science $\rightarrow$ machine learning.
- Customers: occasional / regular $\rightarrow$ finer spending segments.
- Biology: similar expression profiles can be explored at multiple resolutions.

Motivation

A single flat partition may hide useful subgroups. A hierarchy keeps several levels of grouping available.

Real-Life Example: A Music Library

From songs to a hierarchy

Imagine grouping songs using features such as tempo, acoustic energy and instrumental content.

- At a broad level, the hierarchy might separate calmer and more energetic songs.
- Within an energetic group, smaller subgroups could contain fast dance tracks and energetic rock.
- A broad cut gives a few playlists; a finer cut gives more specialized playlists.

Classroom question

Would the same hierarchy appear if we used lyrics and language instead? Likely not: the features define the notion of similarity.

Flat vs. Hierarchical Clustering

Property	Flat clustering	Hierarchical clustering
Output	One partition	Nested sequence of partitions
Examples	K-Means, Fuzzy C-Means	Agglomerative, divisive
Resolution	Selected during fitting	Selected by cutting a hierarchy
Visualization	Scatter plot / memberships	Dendrogram

Key takeaway

At any chosen cut, ordinary agglomerative clustering gives a hard partition.

Agglomerative vs. Divisive

Agglomerative: bottom-up

$$\{A\}, \{B\}, \{C\}, \{D\}$$
$$\downarrow \quad \text{merge}$$
$$\{A, B\}, \{C, D\}$$
$$\downarrow \quad \{A, B, C, D\}$$

Divisive: top-down

$$\{A, B, C, D\}$$
$$\downarrow \quad \text{split}$$
$$\{A, B\}, \{C, D\}$$
$$\downarrow \quad \{A\}, \{B\}, \{C\}, \{D\}$$

Key takeaway

This lecture focuses on agglomerative clustering: start with singletons and merge.

2. Agglomerative Clustering: The Core Idea

- Start with n observations, each in its own cluster.
- Measure distances between all current clusters.
- Merge the pair with the smallest linkage value.
- Update distances involving the new cluster.
- Repeat until one cluster remains, or a stopping rule is met.

Key takeaway

Each merge reduces the number of clusters by one. A full binary hierarchy has $n - 1$ merges.

Algorithm and Stopping Rules

Pseudocode

1. Set $\mathcal{C} = \{\{x_1\}, \ldots, \{x_n\}\}$.
2. Compute the initial pointwise dissimilarities.
3. While the stopping condition has not been reached:
   - Choose a distinct cluster pair minimizing the linkage criterion.
   - Replace that pair by its union; record the merge and its height.
   - Update the relevant cluster distances.
4. Return the hierarchy or a selected partition.

Key takeaway

Stop at K clusters, use a height threshold, or build the full tree first. Standard merges are irreversible.

3. Pointwise Distance vs. Cluster Linkage

Two different questions

$d(x, y)$: How different are two observations?

$D(A, B)$: How different are two clusters?

- For singletons, single/complete/average linkage equal the pointwise distance.
- After merging, a cluster contains multiple observations: a linkage rule is needed.
- Ward uses an increase in SSE; its numerical scale needs separate interpretation.

Euclidean, Manhattan and Minkowski Distance

For $\mathbf{x} = (x_1, \ldots, x_D)$ and $\mathbf{y} = (y_1, \ldots, y_D)$:

$$d_p(\mathbf{x}, \mathbf{y}) = \left(\sum_{j=1}^{D} |x_j - y_j|^p \right)^{1/p}, \qquad p \geq 1.$$

$p = 2$: Euclidean

$$d_2 = \sqrt{\sum_j (x_j - y_j)^2}$$

Straight-line distance.

$p = 1$: Manhattan

$$d_1 = \sum_j |x_j - y_j|$$

Grid / city-block distance.

$p \to \infty$: Chebyshev

$$d_\infty = \max_j |x_j - y_j|$$

Largest coordinate difference.

Distance: A Numerical and Conceptual Example

Let $\mathbf{x} = (1, 2)$ and $\mathbf{y} = (4, 6)$. Then

$$d_2 = \sqrt{3^2 + 4^2} = 5, \quad d_1 = 3 + 4 = 7, \quad d_\infty = 4.$$

Other useful dissimilarities

- Cosine dissimilarity: $1 - \frac{\mathbf{x}^\top \mathbf{y}}{\|\mathbf{x}\|\|\mathbf{y}\|}$ for nonzero vectors; useful for text direction.
- Hamming distance: count or proportion of unequal positions in categorical/binary sequences.
- Precomputed dissimilarities may be used with compatible linkage methods.

Key takeaway

The chosen distance must suit the features and the application. Standard Ward needs Euclidean geometry.

Feature Scaling Changes the Hierarchy

For two observations, suppose the feature differences are

$$\Delta\text{income} = 1000, \qquad \Delta\text{score} = 0.2.$$

Then raw Euclidean distance is

$$\sqrt{1000^2 + 0.2^2} \approx 1000,$$

so income dominates the distance.

Common preprocessing

Standardization: $z_j = (x_j - \bar{x}_j)/s_j$. Min–max: $z_j = (x_j - \min x_j)/(\max x_j - \min x_j)$. Handle missing data and constant features; choose meaningful feature weights.

Key takeaway

Scaling is a modeling decision. Fit preprocessing on the data used for clustering and document it.

Real-Life Example: Scaling Customer Features

Customer segmentation

A shop compares customers by annual spending (rupees) and number of visits. Suppose two customers differ by Rs. 10,000 in spending and by 4 visits.

$$d_{\text{raw}} = \sqrt{10{,}000^2 + 4^2} \approx 10{,}000.$$

The spending difference dominates because its numerical scale is much larger.

What should we do?

Standardize features when differences of one standard deviation should have comparable influence. Use domain weights when spending or visits should matter more. Scaling can change which customers merge first.

4. Linkage: The Critical Transition

Suppose $C_1 = \{A, B\}$ and $C_2 = \{C, D\}$. How should we define $D(C_1, C_2)$?

Choice	Linkage criterion
Closest cross-cluster pair	Single
Farthest cross-cluster pair	Complete
Mean of all cross-cluster distances	Average
Smallest increase in within-cluster SSE	Ward

Key takeaway

Keep the observations and pointwise distances fixed: changing linkage can change the hierarchy.

Single Linkage: Nearest Neighbor

$$\boxed{D_{\text{single}}(A, B) = \min_{x \in A,\, y \in B} d(x, y)}$$

Let $A = \{1, 3\}$ and $B = \{7, 9\}$. The cross-cluster distances are

$$|1 - 7| = 6, \quad |1 - 9| = 8, \quad |3 - 7| = 4, \quad |3 - 9| = 6.$$

$$\boxed{D_{\text{single}}(A, B) = \min(6, 8, 4, 6) = 4}$$

Key takeaway

Only one close cross-cluster pair is needed to make two clusters close. This can recover connected shapes but also causes chaining.

Single Linkage: The Chaining Effect



A chain of nearby points can bridge two dense groups.

- A small link at every step can create one long cluster.
- Endpoints of that cluster can be far apart.
- Bridging noise can connect groups that the application would prefer to separate.

Key takeaway

Single linkage measures connectivity, not the overall diameter of a cluster.

Real-Life Analogy: Single-Linkage Connections

Research collaborators

Consider two groups of researchers. Define $d(x, y)$ as a suitable dissimilarity between two researchers' research-topic vectors.

- Single linkage compares the most similar cross-group pair.
- Two otherwise different groups may merge because one researcher in each group works on a closely related topic.
- A chain of intermediate research interests can join distant areas into one large cluster.

Teaching point

This is an illustration of chaining. The analogy depends on how topic features and dissimilarity are defined; it does not imply the researchers have collaborated directly.

Complete Linkage: Farthest Neighbor

$$\boxed{D_{\text{complete}}(A, B) = \max_{x \in A,\, y \in B} d(x, y)}$$

For $A = \{1, 3\}$ and $B = \{7, 9\}$:

$$\boxed{D_{\text{complete}} = \max(6, 8, 4, 6) = 8}.$$

- Considers the worst-case cross-cluster separation.
- Tends to favor compact groups.
- A single distant observation can strongly influence the criterion.

Key takeaway

Class question: why does one short bridge no longer suffice? The farthest cross-cluster pair still matters.

Real-Life Analogy: Complete-Linkage Teams

Creating compact study groups

Suppose students are represented by a vector of normalized scores on several topics. Let the distance indicate how different their profiles are.

- Complete linkage compares two candidate groups using their most dissimilar cross-group pair.
- Even if one pair has similar profiles, the merge can be delayed when another pair is very different.
- The method tends to form relatively compact groups, but one unusual profile can strongly affect the result.

Classroom question

Would this criterion be suitable if the goal were to mix complementary strengths? A different definition of similarity or a different grouping objective may be needed.

Average Linkage: All Cross-Cluster Pairs

$$\boxed{D_{\text{avg}}(A, B) = \frac{1}{|A||B|} \sum_{x \in A} \sum_{y \in B} d(x, y)}$$

For $A = \{1, 3\}$ and $B = \{7, 9\}$:

$$D_{\text{avg}} = \frac{6 + 8 + 4 + 6}{2 \cdot 2} = \boxed{6}.$$

- Accounts for all cross-cluster distances.
- A compromise between closest-pair and farthest-pair behavior.
- This is pairwise average linkage (UPGMA), not centroid distance.

Key takeaway

The average distance between points is generally different from the distance between averages.

Real-Life Analogy: Average Linkage for Stores

Comparing stores by purchasing patterns

Represent each store by a standardized vector of product-category sales. Consider store groups A and B.

- Compute the distance for every store in A against every store in B.
- Average those cross-group distances to measure the overall separation.
- Unlike single linkage, one similar pair alone does not decide the distance.

Numerical check

If four cross-store distances are $6, 8, 4, 6$, the average-linkage distance is $(6 + 8 + 4 + 6)/4 = 6$. It is not generally the distance between group centroids.

Ward's Method: Minimum Increase in Variance

For a cluster C with centroid μ_C:

$$\mathrm{SSE}(C) = \sum_{x \in C} \|x - \mu_C\|^2, \qquad \mu_C = \frac{1}{|C|} \sum_{x \in C} x.$$

The cost of merging A and B is

$$\boxed{\Delta(A, B) = \mathrm{SSE}(A \cup B) - \mathrm{SSE}(A) - \mathrm{SSE}(B)}.$$

Ward selects a pair minimizing $\Delta(A, B)$.

Key takeaway

Like K-Means, Ward uses within-cluster squared error. Ward optimizes each merge greedily; it does not solve the global K-Means problem.

Real-Life Example: Ward and Delivery Zones

Grouping delivery locations

A company represents nearby delivery locations by projected planar coordinates in kilometers and wants compact zones. Ward compares the increase in within-zone squared distance to the zone center.

$$\Delta(A, B) = \frac{n_A n_B}{n_A + n_B} \|\mu_A - \mu_B\|^2.$$

For groups of sizes 2 and 3, whose centers are 2 km apart:

$$\Delta(A, B) = \frac{2 \cdot 3}{2 + 3}(2)^2 = 4.8 \text{ km}^2.$$

Interpretation

A smaller cost means a smaller increase in squared location spread. Actual routes, road networks and delivery constraints require additional modeling.

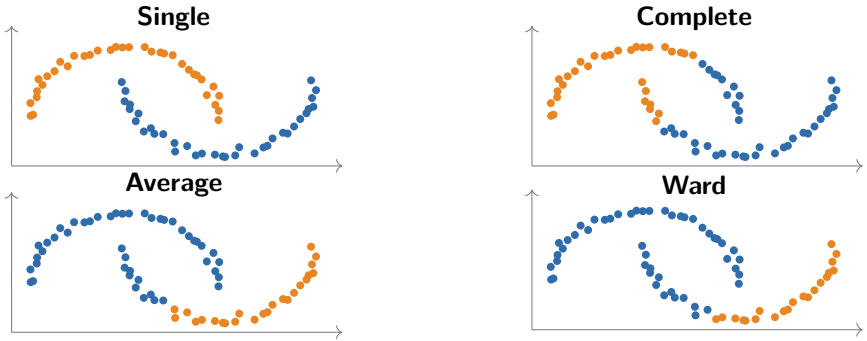
Linkage Criteria: Compare and Check

Linkage	Criterion	Typical tendency	Limitation
Single	Minimum pairwise	Connected / elongated groups	Chaining and noise bridges
Complete	Maximum pairwise	Compact groups	Distant-point sensitivity
Average	Mean pairwise	Overall separation	May blur irregular structures
Ward	Minimum SSE increase	Compact, low-variance groups	Size / shape sensitivity

Quick classroom exercise

If cross-cluster distances are $\{2, 4, 6, 8\}$: single $= 2$, complete $= 8$, average $= 5$. Ward cannot be computed from these three summary values alone.

Visual Comparison: Same 2D Data, Same K=2



Observe the membership changes

All panels use the same unscaled noisy two-moons data and Euclidean geometry. Colors are arbitrary cluster IDs, not known class labels.

Which Linkage Should We Try?

- Connected, elongated structures: single linkage can be useful; inspect bridges and noise.
- Compact groups with controlled worst-case separation: try complete linkage.
- Overall pairwise separation: average linkage offers an intermediate criterion.
- Low-variance Euclidean groups: Ward is a natural candidate.
- Compare several plausible methods using domain meaning, stability and suitable scores.

Teaching prompt

On the two-moons plot, which merges respect connectivity, and which cut across a curved group? No one linkage is best for every dataset.

5. Manual Example: Problem Statement

Perform single-linkage agglomerative clustering with Euclidean distance:

$$A = 1, \qquad B = 2, \qquad C = 4, \qquad D = 8, \qquad E = 9.$$

1. Construct the initial distance matrix.
2. Identify every merge and update the cluster distances.
3. Draw the full dendrogram.
4. Obtain the partition for $K = 2$.

Key takeaway

Tie rule for this worked example: select the alphabetically earliest pair. Reveal and discuss one merge at a time.

Initial Pairwise Distance Matrix

In one dimension, $d(x_i, x_j) = |x_i - x_j|$. For example,

$$d(A, B) = 1, \qquad d(A, C) = 3, \qquad d(B, D) = 6.$$

	A	B	C	D	E
A	0	1	3	7	8
B	1	0	2	6	7
C	3	2	0	4	5
D	7	6	4	0	1
E	8	7	5	1	0

Key takeaway

The matrix is symmetric with a zero diagonal. Start with five singleton clusters. Ignore diagonal entries when selecting a merge.

Merge 1: Resolve the Tie

The minimum off-diagonal distance is 1:

$$d(A, B) = 1, \qquad d(D, E) = 1.$$

Using the stated tie rule:

$$\{A\} + \{B\} \longrightarrow \{A, B\}, \qquad h_1 = 1.$$

Current clusters:

$$\{A, B\}, \quad \{C\}, \quad \{D\}, \quad \{E\}; \qquad K = 4.$$

Key takeaway

Ties can affect binary merge order and sometimes the hierarchy. Here the two tied pairs are disjoint, and both merge at height 1.

Update Distances After Merging A and B

Single linkage uses the closest cross-cluster pair:

$$D(AB, C) = \min(3, 2) = 2,$$
$$D(AB, D) = \min(7, 6) = 6,$$
$$D(AB, E) = \min(8, 7) = 7.$$

	AB	C	D	E
AB	0	2	6	7
C	2	0	4	5
D	6	4	0	1
E	7	5	1	0

Merge 2: D and E

The remaining minimum is $d(D, E) = 1$. Therefore,

$$\{D\} + \{E\} \longrightarrow \{D, E\}, \qquad h_2 = 1.$$

Current clusters are AB, C, and DE; hence $K = 3$.

$$\begin{aligned} D(AB, DE) &= \min(7, 8, 6, 7) = 6, \\ D(C, DE) &= \min(4, 5) = 4, \\ D(AB, C) &= 2. \end{aligned}$$

Key takeaway

Distances between untouched cluster pairs remain unchanged. Only the new cluster needs an updated row and column.

Merge 3: Obtain Two Clusters

	AB	C	DE
AB	0	2	6
C	2	0	4
DE	6	4	0

The smallest distance is $D(AB, C) = 2$:

$$\{A, B\} + \{C\} \longrightarrow \{A, B, C\}, \qquad h_3 = 2.$$

For $K = 2$, the answer is

$$\boxed{C_1 = \{1, 2, 4\}, \qquad C_2 = \{8, 9\}.}$$

Merge 4: Complete the Hierarchy

To construct the full tree, continue from ABC and DE:

$$\begin{aligned} D(ABC, DE) &= \min\{d(A, D), d(A, E), d(B, D), \\ &\qquad d(B, E), d(C, D), d(C, E)\} \\ &= \min(7, 8, 6, 7, 4, 5) = 4. \end{aligned}$$

$$ABC + DE \longrightarrow ABCDE, \qquad h_4 = 4, \quad K = 1.$$

Key takeaway

We can stop early for a partition, or retain all $n - 1$ merges for the complete hierarchy.

Complete Merge Record

Step	Clusters merged	Height	Clusters left
0	None	—	5
1	A, B	1	4
2	D, E	1	3
3	AB, C	2	2
4	ABC, DE	4	1

Result

Heights: $1, 1, 2, 4$. For $K = 2$: $\{A, B, C\}$ and $\{D, E\}$.

What Changes With Complete or Average Linkage?

Pair	Single	Complete	Average
AB, C	2	3	2.5
C, DE	4	5	4.5
AB, DE	6	8	7

All three next merge AB with C. For the final merge:

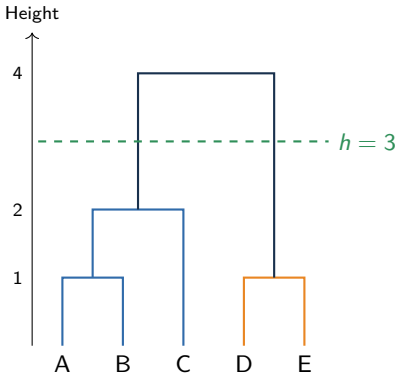
$$D_{\text{avg}}(ABC, DE) = \frac{7+8+6+7+4+5}{6} = \frac{37}{6} \approx 6.167.$$

Method	Full sequence of heights
Single	$1, 1, 2, 4$
Complete	$1, 1, 3, 8$
Average	$1, 1, 2.5, 6.167$

Key takeaway

Same partition here; on other datasets linkage can also change memberships.

6. Dendrogram: The Numerical Example



- Leaves are individual observations.
- Each horizontal branch records a merge.
- The vertical coordinate is the linkage height.
- The cut at $h = 3$ crosses two branches: ABC and DE.

Key takeaway

Height is not generally centroid distance. Rotating branches changes leaf order without changing the hierarchy.

Choosing a Height and a Number of Clusters

Cut height (strictly between merges)	Number of clusters
$0 < h < 1$	5
$1 < h < 2$	3
$2 < h < 4$	2
$h > 4$	1

- At a fixed height, all merges below the cut are retained.
- A height cut cannot give exactly four clusters here: both height-1 merges occur together.
- Stopping after one binary merge can give four clusters, but resolves the tie arbitrarily.

Key takeaway

At exactly a merge height, inclusion depends on the chosen threshold convention. State it or choose a cut between heights.

How Should We Select K?

Three complementary approaches

- Inspect large vertical gaps: cutting below a large jump can be informative.
- Compare candidate cuts with a suitable validation index, such as silhouette.
- Use domain knowledge: the useful resolution depends on the application.

Key takeaway

A large dendrogram gap is a heuristic, not proof of a uniquely correct cluster count. Building the hierarchy and selecting its cut are separate steps.

Real-Life Example: Cutting a Customer Hierarchy

Different decisions need different levels

Imagine a retailer has built one hierarchy of customers using appropriately scaled shopping features.

- A higher cut might give two broad segments for an overall marketing plan.
- A lower cut might produce more detailed segments for tailored campaigns.
- Compare interpretability, segment sizes and validation before choosing a cut.

Key takeaway

A dendrogram can support multiple resolutions. The useful number of groups depends on both the structure and the decision being made.

7. Mathematical Formulation

Let $X = \{x_1, \ldots, x_n\}$ and initially

$$\mathcal{C}^{(0)} = \{\{x_1\}, \ldots, \{x_n\}\}.$$

At iteration t, choose a distinct pair from the current partition:

$$(C_p, C_q) \in \underset{C_i, C_j \in \mathcal{C}^{(t)},\ i \neq j}{\arg\min} D(C_i, C_j).$$

Set $M = C_p \cup C_q$ and update:

$$\boxed{\mathcal{C}^{(t+1)} = (\mathcal{C}^{(t)} \setminus \{C_p, C_q\}) \cup \{M\}.}$$

Key takeaway

The argmin may contain ties. There is no universal global objective shared by all linkages; the procedure makes greedy merge decisions.

Deriving the Single-Linkage Update

Merge disjoint clusters A and B into $M = A \cup B$. For another cluster C,

$$D_{\text{single}}(M, C) = \min_{x \in A \cup B,\ y \in C} d(x, y)$$
$$= \min \left\{ \min_{x \in A, y \in C} d(x, y), \min_{x \in B, y \in C} d(x, y) \right\}.$$

Therefore,

$$\boxed{D_{\text{single}}(A \cup B, C) = \min\{D_{\text{single}}(A, C), D_{\text{single}}(B, C)\}.}$$

Numerical check

If $D(A, C) = 4$ and $D(B, C) = 7$, the updated distance is 4.

Deriving the Complete-Linkage Update

The farthest cross-cluster point must belong to either A or B:

$$D_{\text{complete}}(A \cup B, C) = \max_{x \in A \cup B,\ y \in C} d(x, y)$$
$$= \max\left\{ \max_{x \in A, y \in C} d(x, y), \max_{x \in B, y \in C} d(x, y) \right\}.$$

Thus,

$$\boxed{D_{\text{complete}}(A \cup B, C) = \max\{D_{\text{complete}}(A, C), D_{\text{complete}}(B, C)\}.}$$

Numerical check

For existing complete-linkage distances 4 and 7, the update is 7.

Deriving Average Linkage: Split the Sum

Let $n_A = |A|$, $n_B = |B|$ and $n_C = |C|$. By definition,

$$D_{\text{avg}}(A \cup B, C) = \frac{\sum_{x \in A \cup B} \sum_{y \in C} d(x, y)}{(n_A + n_B) n_C}.$$

Separate the contributions of A and B:

$$= \frac{\sum_{x \in A} \sum_{y \in C} d(x, y) + \sum_{x \in B} \sum_{y \in C} d(x, y)}{(n_A + n_B) n_C}.$$

Since

$$\sum_{x \in A} \sum_{y \in C} d(x, y) = n_A n_C D_{\text{avg}}(A, C),$$

we can express the update using the existing distances.

Deriving Average Linkage: Weighted Update

Substitution gives

$$D_{\text{avg}}(A \cup B, C) = \frac{n_A n_C D_{\text{avg}}(A, C) + n_B n_C D_{\text{avg}}(B, C)}{(n_A + n_B) n_C}.$$

Cancel n_C:

$$\boxed{D_{\text{avg}}(A \cup B, C) = \frac{n_A D_{\text{avg}}(A, C) + n_B D_{\text{avg}}(B, C)}{n_A + n_B}.}$$

For $n_A = 2$, $n_B = 3$, $D(A, C) = 4$, $D(B, C) = 7$:

$$D(A \cup B, C) = \frac{2(4) + 3(7)}{5} = \boxed{5.8}.$$

Key takeaway

Do not use $(4 + 7)/2 = 5.5$ when the cluster sizes differ. Equal weighting of clusters defines a different method (WPGMA).

Ward Derivation I: Centroid and Merge Cost

For disjoint clusters A, B with sizes n_A, n_B,

$$\mu_A = \frac{1}{n_A} \sum_{x \in A} x, \qquad \mu_B = \frac{1}{n_B} \sum_{x \in B} x.$$

Their union $M = A \cup B$ has centroid

$$\boxed{\mu_M = \frac{n_A \mu_A + n_B \mu_B}{n_A + n_B}.}$$

We seek the change in within-cluster squared error:

$$\Delta(A, B) = \mathrm{SSE}(M) - \mathrm{SSE}(A) - \mathrm{SSE}(B), \quad \mathrm{SSE}(A) = \sum_{x \in A} \|x - \mu_A\|^2 .$$

Ward Derivation II: Why the Cross Term Vanishes

Write $x - \mu_M = (x - \mu_A) + (\mu_A - \mu_M)$. Expanding,

$$\sum_{x \in A} \|x - \mu_M\|^2 = \sum_{x \in A} \|x - \mu_A\|^2 + n_A \|\mu_A - \mu_M\|^2 + 2 \left(\sum_{x \in A} (x - \mu_A) \right)^\top (\mu_A - \mu_M).$$

But $\sum_{x \in A} (x - \mu_A) = 0$, so

$$\boxed{\sum_{x \in A} \|x - \mu_M\|^2 = \mathrm{SSE}(A) + n_A \|\mu_A - \mu_M\|^2 .}$$

Applying the same identity to B gives

$$\Delta(A, B) = n_A \|\mu_A - \mu_M\|^2 + n_B \|\mu_B - \mu_M\|^2 .$$

Ward Derivation III: Substitute the Centroid

From the weighted centroid,

$$\mu_A - \mu_M = \frac{n_B}{n_A + n_B}(\mu_A - \mu_B), \qquad \mu_B - \mu_M = -\frac{n_A}{n_A + n_B}(\mu_A - \mu_B).$$

Therefore,

$$\begin{aligned}\Delta(A, B) &= \left[\frac{n_A n_B^2 + n_B n_A^2}{(n_A + n_B)^2}\right] \|\mu_A - \mu_B\|^2 \\ &= \boxed{\frac{n_A n_B}{n_A + n_B} \|\mu_A - \mu_B\|^2}.\end{aligned}$$

Key takeaway

Ward considers both centroid separation and cluster sizes. The cost is nonnegative and has squared feature units.

Ward Numerical: The First Two Merges

For singleton observations $A = 1, B = 2, C = 4, D = 8, E = 9$:

$$\Delta(A, B) = \tfrac{1}{2}(1 - 2)^2 = 0.5, \qquad \Delta(D, E) = 0.5.$$

The next-smallest initial cost is $\Delta(B, C) = \tfrac{1}{2}(2 - 4)^2 = 2$. After A, B merge, $\mu_{AB} = 1.5$ and

$$\Delta(AB, C) = \frac{2 \cdot 1}{3}(1.5 - 4)^2 = \frac{25}{6} \approx 4.167.$$

The unmerged D, E pair still costs 0.5, so it merges next.

Key takeaway

Recompute costs after every merge; the second merge is D, E, before AB, C.

Ward Numerical: Compare All Three Candidates

After the first two merges, clusters are AB, C, DE:

$$\mu_{AB} = 1.5, \quad \mu_C = 4, \quad \mu_{DE} = 8.5.$$

$$\Delta(AB, C) = \tfrac{2}{3}(1.5 - 4)^2 = 25/6 \approx 4.167,$$
$$\Delta(C, DE) = \tfrac{2}{3}(4 - 8.5)^2 = 13.5,$$
$$\Delta(AB, DE) = \tfrac{2 \cdot 2}{4}(1.5 - 8.5)^2 = 49.$$

Choose $AB + C$. Now $\mu_{ABC} = 7/3$ and

$$\Delta(ABC, DE) = \frac{3 \cdot 2}{5}\left(\frac{7}{3} - \frac{17}{2}\right)^2 = \frac{1369}{30} \approx 45.633.$$

Ward: SSE Costs vs. Dendrogram Heights

Merge	SSE increase Δ	SciPy height $h = \sqrt{2\Delta}$
A, B	0.5	1
D, E	0.5	1
AB, C	$25/6 \approx 4.167$	$\sqrt{25/3} \approx 2.887$
ABC, DE	$1369/30 \approx 45.633$	$\sqrt{1369/15} \approx 9.553$

Total SSE telescopes across the merges:

$$0.5 + 0.5 + 25/6 + 1369/30 = 50.8 = \sum_{x \in X} (x - 4.8)^2.$$

Key takeaway

Do not label Δ as the plotted software height without checking its convention. Feed Euclidean observations to standard Ward, not arbitrarily squared input distances.

Optional: Ward Distance Update

Under the convention $h(A,B) = \sqrt{2\Delta(A,B)}$, let $M = A \cup B$ and $T = n_A + n_B + n_C$. Then

$$\boxed{h(M,C) = \sqrt{\frac{n_A + n_C}{T} h(A,C)^2 + \frac{n_B + n_C}{T} h(B,C)^2 - \frac{n_C}{T} h(A,B)^2}.}$$

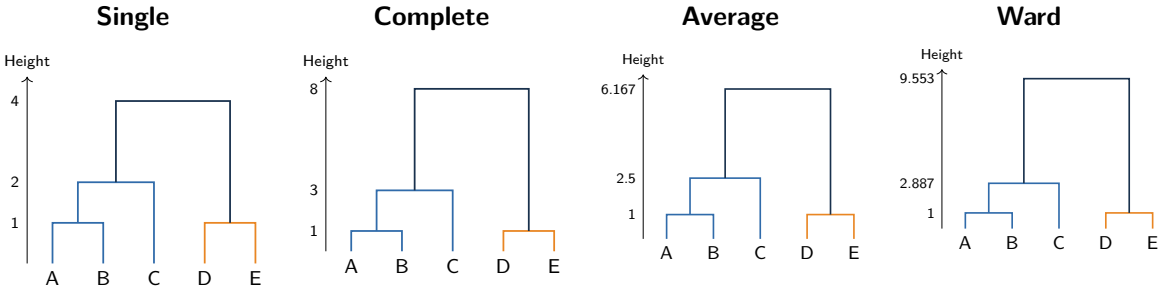
This follows by substituting the weighted centroid into the merge-cost formula and using the squared-distance identity

$$\|\mu_M - \mu_C\|^2 = \frac{n_A}{n_A + n_B} \|\mu_A - \mu_C\|^2 + \frac{n_B}{n_A + n_B} \|\mu_B - \mu_C\|^2 - \frac{n_A n_B}{(n_A + n_B)^2} \|\mu_A - \mu_B\|^2 .$$

Key takeaway

An advanced implementation detail: the centroid-based Δ formula is sufficient for manual Ward calculations.

Four Dendrograms: Same Data, Different Heights



Each panel uses its own vertical scale; read the tick values. Ward shows $\sqrt{2\Delta}$.

Key takeaway

The topology happens to agree for these five points. Heights and the interpretation of linkage still differ.

8. Silhouette Score: Definition

For observation i in a cluster containing at least two points,

$$a(i) = \frac{1}{|C_i| - 1} \sum_{j \in C_i, j \neq i} d(i,j),$$

$$b(i) = \min_{C \neq C_i} \frac{1}{|C|} \sum_{j \in C} d(i,j), \qquad \boxed{s(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}}}.$$

- $s \approx 1$: good separation; $s \approx 0$: similar within/between distances.
- $s < 0$: the point is, on average, closer to another group.
- Use the mean across observations to compare cuts; singleton score is conventionally 0.

Key takeaway

Evaluation requires $2 \leq K \leq n - 1$. If $a = b = 0$, use the conventional score 0.

Silhouette Score: Worked Example

Use $C_1 = \{1, 2, 4\}$ and $C_2 = \{8, 9\}$. For $x = 4$,

$$a(4) = \frac{|4-1| + |4-2|}{2} = \frac{3+2}{2} = 2.5,$$

$$b(4) = \frac{|4-8| + |4-9|}{2} = \frac{4+5}{2} = 4.5.$$

Hence

$$\boxed{s(4) = \frac{4.5 - 2.5}{4.5} = \frac{4}{9} \approx 0.444.}$$

Key takeaway

This is one observation's score, not the mean score for the whole partition. A positive score supports its assignment under this distance.

Real-Life Example: Interpreting Silhouette

Grouping customers by shopping behavior

After scaling features, consider a customer assigned to a segment. Suppose the mean distance to the other members of that segment is $a = 2$, and the smallest mean distance to a different segment is $b = 5$.

$$s = \frac{5 - 2}{\max(2, 5)} = 0.6.$$

- The positive score suggests this customer is closer on average to their assigned group.
- A negative score would suggest another group is closer on average.
- Inspect the mean and individual scores; also check whether the groups make business sense.

Davies–Bouldin Index

For Euclidean features, one common definition uses

$$S_i = \frac{1}{|C_i|} \sum_{x \in C_i} \|x - \mu_i\|, \qquad M_{ij} = \|\mu_i - \mu_j\|.$$

$$\boxed{DB = \frac{1}{K} \sum_{i=1}^{K} \max_{j \neq i} \frac{S_i + S_j}{M_{ij}}.}$$

- Within-cluster scatter should be small relative to center separation.
- Lower values generally indicate more compact and separated groups.
- Identical centers need special handling; centroid-based scoring can favor compact shapes.

Key takeaway

Indices support comparison; they do not establish a universally correct clustering. Use metrics consistent with the representation and goal.

Advantages and Limitations

Advantages

- Interpretable nested structure.
- Several possible cuts from one tree.
- Multiple linkage and distance choices.
- No randomly initialized centroids.
- Useful for exploratory analysis.

Limitations

- Greedy, irreversible merges.
- Scaling and distance sensitivity.
- Outliers and noise can distort results.
- Tie rules can matter.
- Dense pairwise storage can be expensive.

Key takeaway

No centroid initialization does not imply a unique answer: ties and implementation choices can still affect a hierarchy.

Computational Complexity

For n observations there are

$$\boxed{\frac{n(n-1)}{2}}$$

distinct off-diagonal pairwise distances.

- Dense pairwise storage takes $O(n^2)$ memory.
- Computing Euclidean distances for D features costs $O(n^2 D)$.
- A naive full scan at each merge can lead to $O(n^3)$ time.
- Optimized common methods can use $O(n^2)$ time; details depend on the implementation.
- Sparse connectivity or specialized algorithms may change memory and runtime behavior.

Key takeaway

At $n = 100{,}000$, even a condensed float64 distance vector is about 40 GB (decimal), before working memory.

K-Means vs. Fuzzy C-Means vs. Agglomerative

Property	K-Means	Fuzzy C-Means	Agglomerative
Structure	Flat	Flat	Hierarchical
Membership	Hard	Soft degrees	Hard at a cut
Cluster count	Choose K	Choose C	Start with n; choose a cut
Initialization	Centers / seed	Centers / memberships	Singleton clusters
Operation	Assign, update centers	Update memberships, centers	Merge clusters
Output	Labels and centers	Memberships and centers	Tree and partitions
Reassignment	Possible each iteration	Memberships change	Prior merges retained
Criterion	Within-cluster SSE	Fuzzy weighted SSE	Depends on linkage

Key takeaway

K-Means: which center? FCM: how much membership? Agglomerative: which pair should merge?

Applications: Representation, Distance, Interpretation

Application	Representation	Distance / method	Interpretation
Documents	TF–IDF or embeddings	Cosine with compatible linkage	Broad topics and subtopics
Bioinformatics	Gene-expression profiles	Scaled Euclidean or suitable dissimilarity	Related expression patterns
Customers	Frequency, spending, recency	Scale features; compare linkages	Broad segments and subsegments

Key takeaway

Specify what similarity means before interpreting a group. A clustering hierarchy alone does not establish biological ancestry or a causal relationship.

Real-Life Example: Outliers and Noise

Customer spending

Suppose most customers spend between Rs. 1,000 and Rs. 20,000, but one wholesale buyer spends Rs. 500,000.

- The extreme observation can strongly influence distances and some merge decisions.
- Check whether it is a data-entry error or a valid but different type of customer.
- Consider suitable scaling or a separate analysis, and document the reason.

Classroom question

Should the wholesale buyer automatically be deleted? No: first investigate whether the observation is erroneous or meaningful to the task.

Common Mistakes to Prevent

- Using centroid distance as if it were single, complete or average linkage.
- Averaging two average-linkage distances without cluster-size weights.
- Choosing the minimum diagonal zero instead of a distinct cluster pair.
- Reading horizontal leaf spacing as a distance measurement.
- Equating Ward SSE increase with every plotted Ward height.
- Ignoring scaling, ties, outliers or the selected distance.
- Treating numeric cluster IDs as ranked or meaningful labels.
- Claiming greedy merges guarantee a globally optimal partition.

9. Summary: From Data to Final Clusters

$$\text{Data} \longrightarrow \text{Distances} \longrightarrow \text{Linkage} \longrightarrow \text{Merges}$$
$$\longrightarrow \text{Dendrogram} \longrightarrow \text{Chosen cut and clusters.}$$

- Bottom-up: each observation starts as a singleton.
- A full tree records $n - 1$ merges.
- Single = nearest pair; complete = farthest pair; average = all pairs.
- Ward = smallest increase in within-cluster SSE at each step.
- Use the hierarchy, validation and domain knowledge to choose a cut.
- Scaling and linkage determine what the resulting groups mean.

Viva Questions: Concepts

1. What is hierarchical clustering? How do agglomerative and divisive differ?
2. Why is agglomerative clustering called bottom-up?
3. What is a proximity matrix? Which entries represent candidate merges?
4. How do pointwise distance and linkage distance differ?
5. Explain single, complete, average and Ward linkage.
6. What is chaining? How can noise create a bridge?
7. Why does feature scaling matter?
8. How is a dendrogram constructed and read?

Viva Questions: Mathematics and Interpretation

9. Can one dendrogram provide different numbers of clusters?
10. Why are prior merges irreversible in the standard algorithm?
11. Derive the size-weighted average-linkage update.
12. Derive Ward's merge cost and explain the zero cross term.
13. How do Ward's SSE cost and SciPy's plotted height differ?
14. Compare agglomerative clustering with K-Means and FCM.
15. What are the time and memory limitations?
16. How can ties prevent a height cut from producing every integer K?

Student Assignment

Use Euclidean distance on

$$X = \{2, 5, 6, 10, 15, 18\}.$$

1. Construct the initial distance matrix.
2. Solve using single linkage, then complete linkage.
3. Record all merge pairs and heights; state a deterministic tie rule.
4. Draw both dendrograms and identify each $K = 2$ partition.
5. Explain differences in membership and heights.
6. Optional: compute a silhouette score for one observation in a two-cluster result.

Key takeaway

Show intermediate matrices or distance updates, not only the final partition.

Suggested Two-Session Teaching Plan

Session	Time	Topics
1	10 min	Recap, motivation and hierarchy
1	15 min	Algorithm, distance and scaling
1	20 min	Linkage criteria and visual intuition
1	15 min	Begin manual example
2	15 min	Finish example and dendrogram cuts
2	20 min	Linkage updates and Ward derivation
2	10 min	Evaluation, comparison and limitations
2	10 min	Real-life applications and interpretation
2	5 min	Summary, viva and assignment

Key takeaway

Use the optional Ward update and extended questions as reference material if the classroom discussion needs more time.

References and Further Reading

- Ward, J. H. (1963). Hierarchical Grouping to Optimize an Objective Function. *Journal of the American Statistical Association*, 58(301), 236–244.
- Rousseeuw, P. J. (1987). Silhouettes: A Graphical Aid to the Interpretation and Validation of Cluster Analysis. *Journal of Computational and Applied Mathematics*, 20, 53–65.
- SciPy: linkage documentation; fcluster documentation.
- scikit-learn: hierarchical clustering user guide; silhouette score.
- Design reference: supplied *K-Means and Fuzzy C-Means Clustering* lecture PDF.

Reproducibility: all manual examples use the stated distances and tie rule. Real-life scenarios are illustrative examples.